

On 180M voxels Network still fails ADR 0012 by one strand

Stage / surface	What is counted	Residual	Of	Reading
Energy scale-index	mismatched voxels	0	16.8M	exact discrete field
Energy float	max $ \Delta $	$\sim 2e-11$	allclose	library FP drift only
Vertices	position / scale	0	exact	bit-identical geometry
Edges ownership	disagreed voxels	~ 8	5.84M claimed	99.9999% map agree
Edges connections	pair multiset Δ	1	69,500	one equal-metric swap
Network strands	strand multiset Δ	1	48,049	ADR 0012 FAIL (open gate)

Takeaway: Energy/Vertices closed; Edges ownership/count evaluated PASS on canonical_full_v16.
Open ship gate: Network strand multiset FAIL (48,048 / 48,049) — one degree-pruning pair swap
(crop: MATLAB [4212, 6281] vs Python [4043, 6281]). Not an independent Network bug.